

# Ajinkya A. Ingale

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## PROFESSIONAL SUMMARY

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Versatile and results driven Machine Learning Engineer with over 10 years of experience delivering end-to-end ML / AI solutions across industries including finance, real estate, and augmented reality. Possess a strong background in building scalable systems, recommendation engines, and intelligent automation. Adept at translating research into production-ready solutions to create high-impact, reliable AI applications.

## TECHNICAL SKILLS

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- **Proficient:** Python, Data Science, Machine Learning Engineering, Recommendation Systems, Natural Language Processing, **LLM**(Retrieval Augmented Generation), Neural Networks, Pytorch **Tensorflow**, **Keras**, **OpenCv**, **DI4J**, Flask, Kube-flow, Docker, **GCP**, **Dataproc**, **Spark**, **Airflow**, Kubernetes, BigQuery, CI/CD Jenkins & Terraform, Google AI Platform, VertexAI, BigQuery, Google Cloud Storage
- **Exposure:**, R, Scala, Unity3D, Jenkins, Neuro-symbolic AI

## EXPERIENCE at University of California, Merced

May 2024- present

- **Developed a robust LLM evaluation framework** in Python using LangChain, OpenAI API, and custom logic modules to assess model performance on spatial reasoning, symbolic manipulation, and logic-based tasks. Enabled scalable testing infrastructure for analyzing model behavior across diverse reasoning scenarios.
- **Created benchmark datasets and modular prompt templates** to measure consistency, accuracy, and rule-following in large language models. Evaluations supported fine-tuning, prompt optimization, and safety alignment across multiple model iterations.
- **Exposed critical failure modes** such as inconsistent state tracking and symbolic reasoning drift. Findings directly informed improvements to model interpretability, robustness, and system behavior in high-stakes environments.
- **Contributed actionable insights toward improving LLM alignment, reliability, and UX** by focusing on structured reasoning use cases in real-world applications supporting teams working on truthfulness, safety, and generalization.

## WORK EXPERIENCE

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### CoreLogic | Sr. Machine Learning Engineer

Feb 2022 - April 2024

- **Built an automated ML training and validation pipeline** using GCP (Cloud Functions, Pub/Sub, BigQuery, Airflow, Dataproc), automating data ingestion, preprocessing, and model retraining. **Reduced retraining costs and standardized training/inference workflows for scalability.**
- **Developed a RAG-based Question-Answering system** using Hugging Face and a Vector Database to support domain-specific risk analysis. **Improved contextual accuracy and reliability of automated risk-profile calculations.**
- **Implemented CI/CD automation for ML pipelines** using Jenkins, Terraform, and Cloud Run, enabling automated provisioning, testing, and deployment. **Decreased deployment time by 25% and improved release stability and reproducibility.**

### ChewedPixelStudios | Sr. Machine Learning Engineer

Feb 2020 - Jan 2022

- **Designed architecture for a scalable ML systems and workflows** by evaluating cloud-native components, data pipelines, and model-serving patterns, ensuring reliability, reproducibility, and seamless deployment across environments core to MLOps best practices.

- **Developed end-to-end object detection pipelines** for AR applications using Python, IBM Watson, OpenCV, TensorFlow, and Keras, automating dataset preparation, annotation, augmentation, training, and validation to support continuous model improvement.
- **Integrated trained ML models into Unity3D AR environments** and established deployment workflows enabling real-time inference, performance monitoring, and version control aligned with ML lifecycle management and **delivered production-ready ML pipelines with automated processes.**

#### **Wells Fargo | Natural Language Processing Engineer | San Francisco**

**Jan 2018 - Jan 2020**

- **Migrated a conversational AI system** from regex-based logic to scalable deep learning models using Java and Bidirectional LSTMs, enabling more robust intent understanding and **increasing user engagement by 30%.**
- **Trained and optimized an ensemble of intent and entity classification models** using DL4J, OpenNLP, Python, Scikit-learn, NLTK, TensorFlow, and word vector embeddings, **improving accuracy by 25%** and **reducing response latency by 65%** compared to benchmark baselines.
- **Implemented automated ML deployment workflows** using Kafka, Kubernetes, and Docker to streamline model delivery, monitoring, and versioning aligned with MLOps best practices, **reduced commit-to-deploy time by 60%.**
- **Established scalable, production-ready model training and inference pipelines**, integrating data processing, model evaluation, and continuous optimization to support reliable real-time conversational AI capabilities.
- **Built a Voice Authentication system using Gaussian Mixture Models (GMMs)** for speaker verification, implementing feature extraction (MFCCs), model training, and real-time inference pipelines, improving authentication accuracy and enabling secure user identity validation.

#### **Financial Engines | Machine Learning Engineer | Sunnyvale**

**Dec 2013 - Nov 2017**

- **Designed and productionized topic modeling pipelines** using Java, Apache UIMA, Mahout, and Stanford NLP to classify product complaints and system issues. Deployed REST-based services with monitoring hooks, **reducing resolution time by 65% and while achieving 25% cost savings over projections..**
- **Implemented real-time anomaly detection systems** using Elasticsearch, Splunk, SQL, Lucene, TF-IDF, and N-gram modeling for automated log analysis and fault detection. **Decreased system failure rates by 23%** and improved incident response through alerting and dashboard integration.
- **Built and maintained automated CI/CD pipelines** using Jenkins, Docker, and Kubernetes for ML services and data workflows, ensuring reproducible builds, version-controlled deployments, and **a 60% reduction in commit-to-production time.**
- **Contributed to scalable, fault-tolerant ETL systems** using Java, SQL, REST APIs, and Jenkins to support downstream ML and analytics workloads. Focused on data reliability, recovery mechanisms, and on-call readiness aligned with SRE principles.
- **Developed observability features across ML pipelines** including metric logging, model performance monitoring, and error tracing to support continuous delivery and real-time diagnostics in production environments.

## **EDUCATION**

Masters in Cognitive and Information Sciences, University of California, Merced

December 2025

Masters in Software Engineering, *San Jose State University*

December 2013

Bachelors in Engineering, Information Technology, *University of Pune*

May 2008